The maximum total across all homework assignments this semester is 50 points, corresponding to a bonus of 0.5 on the tutorial grade.

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Notes:

- Throughout, N, k, and d are integers at least 2, and t is a real number.
- For positive integers m, n, $\mathcal{M}_{m,n}(\mathbb{R})$ denotes the real $m \times n$ matrices.
- For brevity, we use $[1, N]$, to mean $\{1, \ldots, N\}$.
- For a matrix A, A^T denotes its transpose.
- All random objects are defined on the finite probability space $(\Omega, \mathcal{A}, \mathbb{P})$.
- For a Euclidean space E, such as $\mathbb{R}^n$, write $\langle \cdot, \cdot \rangle$ and $\| \cdot \|$ for its inner product and induced norm.

Problem statement and objective

What is the largest number of points in $\mathbb{R}^n$ that can all be at equal distances from one another? In the plane ($n = 2$) the answer is 3, forming an equilateral triangle; for $n = 3$ it is 4, forming a regular tetrahedron. One can check that in $\mathbb{R}^n$ it is $n + 1$.

Thus, for example, 50 vectors in $\mathbb{R}^{100}$ can be mutually equidistant. The preceding observation shows that depicting their positions in two dimensions is impossible without changing the distances! This is a relatively simple manifestation of the *curse of dimensionality*, a collective term for the often counterintuitive properties of high-dimensional spaces.

This assignment asks:

> Can N points in a high-dimensional Euclidean space be linearly projected into a lower-dimensional space without substantially distorting their pairwise distances?

More precisely, let $v_1, \ldots, v_N \in \mathbb{R}^d$ be distinct. For $\varepsilon \in (0, 1)$, call a linear map $f : \mathbb{R}^d \to \mathbb{R}^k$ an *ε-isometry* for these vectors if

$$\forall (i, j) \in [1, N]^2 : \quad (1 - \varepsilon) \|v_i - v_j\| \leq \|f(v_i) - f(v_j)\| \leq (1 + \varepsilon) \|v_i - v_j\|.$$

Objective: For which k does there exist a linear ε-isometry $f : \mathbb{R}^d \to \mathbb{R}^k$ for $v_1, \ldots, v_N$?

We will prove the following theorem, first established by William B. Johnson and Joram Lindenstrauss in 1984:

> *There is an absolute constant $c > 0$ such that, for all integers $N, d \geq 2$ and every set of N distinct vectors in $\mathbb{R}^d$, the condition*
>
> $$k \geq \frac{c \ln(N)}{\varepsilon^2},$$
>
> *is sufficient for an ε-isometry $f : \mathbb{R}^d \to \mathbb{R}^k$ for $v_1, \ldots, v_N$ to exist.*

The only known proofs of this theorem use probabilistic methods. Section 1 develops preliminary results on convexity and probability. Section 2 proves a concentration result of Talagrand, which Section 3 applies to the Johnson–Lindenstrauss theorem.

1 Preliminary results

1.1 Projection onto a closed convex set

Let E be a Euclidean space.

1. Let $a, b \in E$. Prove and geometrically interpret
$$\|a+b\|^2 + \|a-b\|^2 = 2\left(\|a\|^2 + \|b\|^2\right).$$

2. Deduce that if $u, v, v' \in E$, $v \neq v'$, and $\|u-v\| = \|u-v'\|$, then
$$\left\|u - \frac{v+v'}{2}\right\| < \|u-v\|.$$

3. Let $F \subset E$ be nonempty and closed, and $u \in E$. Prove that there exists $v \in F$ such that
$$\forall w \in F: \quad \|u-v\| \leq \|u-w\|.$$

4. Deduce that if $C \subset E$ is nonempty, closed and convex, then for every $u \in E$ there is a unique $v \in C$ such that
$$\forall w \in C: \quad \|u-v\| \leq \|u-w\|.$$
Call v the *projection* of u onto C, and write $d(u, C) := \|u-v\|$.

1.2 Hölder's inequality for expectations

Let p, q be positive real numbers satisfying
$$\frac{1}{p} + \frac{1}{q} = 1.$$

1. Use concavity of the logarithm to prove, for positive a, b,
$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

2. Deduce that random variables X, Y on a finite probability space satisfy
$$\mathbb{E}\left[|XY|\right] \leq \left(\mathbb{E}\left[|X|^p\right]\right)^{1/p} \left(\mathbb{E}\left[|Y|^q\right]\right)^{1/q}.$$
(*First consider* $\mathbb{E}\left[|X|^p\right] = \mathbb{E}\left[|Y|^q\right] = 1$.)

1.3 Random variables with sub-Gaussian tails

Let $X : \Omega \to \mathbb{R}$ and suppose there are $a, b > 0$ such that, for all $t > 0$,
$$\mathbb{P}(|X| \geq t) \leq a \exp(-bt^2).$$

1. Prove that
$$\mathbb{E}\left[X^2\right] = 2\int_0^{+\infty} t\,\mathbb{P}(|X| \geq t)\,\mathrm{d}t.$$

2. Show that $\mathbb{E}\left[X^2\right] \leq a/b$.

3. For δ with $\delta^2 \leq a/b$, prove
$$\mathbb{P}(|X+\delta| \geq t) \leq \mathbb{P}(|X| \geq t - |\delta|).$$

4. Prove that
$$-b(t-|\delta|)^2 \leq a - \frac{1}{2}bt^2.$$

5. Deduce that for $t \geq |\delta|$,
$$\mathbb{P}(|X+\delta| \geq t) \leq a\exp(a)\exp\Big(-\frac{1}{2}bt^2\Big).$$

6. Show that this inequality also holds for $0 \leq t < |\delta|$.

2 Talagrand's inequality

Let E be an n-dimensional Euclidean space, $n \geq 1$, with orthonormal basis $(e_1, \ldots, e_n)$. Let $\varepsilon_1, \ldots, \varepsilon_n$ be independent Rademacher variables, meaning $\mathbb{P}(\epsilon_1 = 1) = \mathbb{P}(\epsilon_1 = -1) = 1/2$, and set

$$X := \sum_{i=1}^{n} \varepsilon_i \, e_i.$$

Our aim is to show that for every nonempty closed convex $C \subset E$,

$$\mathbb{P}\,(X \in C)\ \mathbb{E}\left[\exp\left(\frac{1}{8}\, d(X, C)^2\right)\right] \leq 1, \tag{1}$$

and use this to establish Talagrand's inequality (2).

2.1 Two special cases

1. Consider the case where C does not intersect
$$X(\Omega) := \{x \in E : X(\omega) = x \text{ for some } \omega \in \Omega\},$$
that is, $C \cap X(\Omega) = \emptyset$.
For the remaining parts of this subsection, assume $C \cap X(\Omega)$ consists of a single vector u.
2. Show that $\frac{1}{4} d(X, u)^2 \sim \mathrm{Bin}(n, 1/2)$.
3. Find the expectation of $\exp(\frac{1}{8} d(X, u)^2)$ and show that it is at most 2^n.
4. Prove $d(X, C) \leq d(X, u)$, and establish (1).

2.2 Induction

Now suppose $C \subset E$ is closed and convex, with at least two elements in $C \cap X(\Omega)$. Without loss of generality, assume two of those vectors differ in their last coordinate. We prove (1) by induction on $n = \dim E$.

1. Consider $n = 1$.

2.3 Properties of C_{+1} and C_{-1}

Let

$$E' := \mathrm{span}(e_1, \ldots, e_{n-1}), \quad \pi : E \to E' \text{ be the orthogonal projection}, \quad X' := \pi(X) = \sum_{i=1}^{n-1} \varepsilon_i \, e_i.$$

For $t \in \{-1, 1\}$, let

$$H_t := E' + t\, e_n, \quad C_t := \pi\big(C \cap H_t\big).$$

1. Show that for every $x' \in E'$ and $t \in \{-1, 1\}$,
$$x' \in C_t \iff x' + t\, e_n \in C.$$
2. Prove that C_{+1}, C_{-1} are nonempty closed convex subsets of E'.
3. For $t \in \{-1, 1\}$, let Y_t be the projection of X' onto C_t. Show that
$$\mathbb{P}(X \in C) = \frac{1}{2}\, \mathbb{P}(X' \in C_{+1}) + \frac{1}{2}\, \mathbb{P}(X' \in C_{-1}).$$

2.4 A key inequality

Let $0 \leq \lambda \leq 1$.

1. Show that
$$d(X, C) \leq \Big\| (1 - \lambda)\big(Y_{\varepsilon_n} + \varepsilon_n e_n\big) + \lambda\big(Y_{-\varepsilon_n} - \varepsilon_n e_n\big) - X \Big\|.$$
2. Then prove that
$$d(X, C)^2 \leq 4\lambda^2 + \Big\| (1 - \lambda)(Y_{\varepsilon_n} - X') + \lambda(Y_{-\varepsilon_n} - X') \Big\|^2$$
and
$$d(X, C)^2 \leq 4\lambda^2 + (1 - \lambda)\|Y_{\varepsilon_n} - X'\|^2 + \lambda \|Y_{-\varepsilon_n} - X'\|^2,$$
that is,
$$d(X, C)^2 \leq 4\lambda^2 + (1 - \lambda) d(X', C_{\varepsilon_n})^2 + \lambda\, d(X', C_{-\varepsilon_n})^2.$$

2.5 Conditional expectations

Let

$$p_+ := \mathbb{P}(X' \in C_{+1}), \quad p_- := \mathbb{P}(X' \in C_{-1}),$$

and assume without loss of generality that $p_+ \geq p_-$.

1. Show that $p_- > 0$.
2. Prove that for every $\lambda \in [0,1]$,

$$\mathbb{E}\left[\exp\left(\frac{1}{8}\,d(X,C)^2\right) \mid \varepsilon_n = -1\right] \leq \exp\left(\frac{\lambda^2}{2}\right)\ \mathbb{E}\left[\left(\exp\left(\frac{1}{8}\,d(X',C_{-1})^2\right)\right)^{1-\lambda}\ \left(\exp\left(\frac{1}{8}\,d(X',C_{+1})^2\right)\right)^{\lambda}\right].$$

3. Deduce that

$$\mathbb{E}\left[\exp\left(\frac{1}{8}\,d(X,C)^2\right) \mid \varepsilon_n = -1\right] \leq \exp\left(\frac{\lambda^2}{2}\right)\ \mathbb{E}\left[\exp\left(\frac{1}{8}\,d(X',C_{-1})^2\right)\right]^{1-\lambda}\ \mathbb{E}\left[\exp\left(\frac{1}{8}\,d(X',C_{+1})^2\right)\right]^{\lambda}.$$

4. Using the induction hypothesis, justify

$$\mathbb{E}\left[\exp\left(\frac{1}{8}\,d(X,C)^2\right) \mid \varepsilon_n = 1\right] \leq \frac{1}{p_+}.$$

and then show that, for every $\lambda \in [0,1]$,

$$\mathbb{E}\left[\exp\left(\frac{1}{8}\,d(X,C)^2\right)\right] \leq \frac{1}{2}\ \left(\frac{1}{p_+} + \exp\left(\frac{\lambda^2}{2}\right)\ \frac{1}{(p_-)^{1-\lambda}}\ \frac{1}{(p_+)^{\lambda}}\right).$$

2.6 Optimization

1. Let $\lambda = 1 - p_-/p_+$. Show that

$$\mathbb{E}\left[\exp\left(\frac{1}{8}\,d(X,C)^2\right)\right] \leq \frac{1}{2p_+}\ \left(1 + \exp\left(\frac{\lambda^2}{2}\right)\ (1-\lambda)^{\lambda-1}\right).$$

2. Prove that for every $x \in [0,1)$,

$$\frac{x^2}{2} + (x-1)\,\ln(1-x)\ \leq\ \ln(2+x) - \ln(2-x).$$

3. Deduce that for every $x \in [0,1)$,

$$1 + \exp\left(\frac{x^2}{2}\right)\ (1-x)^{x-1}\ \leq\ \frac{4}{2-x}.$$

4. Complete the proof of (1).
5. Prove Talagrand's inequality:

> *For every nonempty closed convex $C \subset E$ and every $t > 0$,*
>
> $$\mathbb{P}(X \in C)\,\mathbb{P}(d(X,C) \geq t)\ \leq\ \exp\left(-\frac{t^2}{8}\right). \tag{2}$$

3 The Johnson–Lindenstrauss theorem

In this section consider

$$E = \mathcal{M}_{k,d}(\mathbb{R}),$$

equipped with the inner product $\langle A, B\rangle_F := \operatorname{tr}(A^T B)$ and its induced (Frobenius) norm $\|\cdot\|_F$. Identify $\mathbb{R}^d$ with $\mathcal{M}_{d,1}(\mathbb{R})$, viewing vectors as column matrices. Let $u \in \mathbb{R}^d$, $\|u\| = 1$, and define

$$g(M) := \|M\,u\|, \quad M \in \mathcal{M}_{k,d}(\mathbb{R}).$$

Let

$$X = (\varepsilon_{ij})_{1 \leq i \leq k,\ 1 \leq j \leq d}$$

be a random matrix in $\mathcal{M}_{k,d}(\mathbb{R})$ whose entries ε_{ij} are independent Rademacher variables.

3.1 A concentration inequality

1. Show that
$$C = \{M \mid g(M) \leq r\}$$
is a closed convex subset of $\mathcal{M}_{k,d}(\mathbb{R})$.

2. Prove that every matrix M satisfies
$$\|M\,u\| \leq \|M\|_F.$$

3. Let $r, t \in \mathbb{R}$, $t > 0$. Show that $d(M, C) < t$ implies $g(M) < r + t$.

4. Deduce that
$$\mathbb{P}(g(X) \leq r)\,\mathbb{P}(g(X) \geq r + t) \;\leq\; \exp\left(-\frac{t^2}{8}\right).$$

3.2 Medians

1. Explain why $g(X)$ has at least one median.

2. Prove that for every $t > 0$,
$$\mathbb{P}\left(|g(X) - m| \geq t\right) \;\leq\; 4\,\exp\left(-\frac{t^2}{8}\right),$$
and that
$$\mathbb{E}\left[(g(X) - m)^2\right] \leq 32.$$

3. Show that $\mathbb{E}\left[g(X)^2\right] = k$ and $\mathbb{E}\left[g(X)\right] \leq \sqrt{k}$.

4. Deduce that
$$(\sqrt{k} - m)^2 \leq \mathbb{E}\left[(g(X) - m)^2\right].$$

3.3 A key lemma

1. Show that for every $t > 0$,
$$\mathbb{P}\left(|g(X) - \sqrt{k}| \geq t\right) \;\leq\; 4\,\exp(4)\,\exp\left(-\frac{t^2}{16}\right).$$

2. Define $A_k = X/\sqrt{k}$, and let $\varepsilon \in (0,1)$, $\delta \in (0, 1/2)$, with
$$k \geq \frac{160\,\ln(1/\delta)}{\varepsilon^2}.$$
Prove that for every vector u of unit norm,
$$\mathbb{P}\left(\big|\,\|A_k\,u\| - 1\big| > \varepsilon\right) < \delta.$$

3.4 Conclusion

Let $v_1, \ldots, v_N$ be distinct vectors in $\mathbb{R}^d$. For $(i,j) \in [1,N]^2$, $i < j$, define
$$E_{ij} := \left((1-\varepsilon)\,\|v_i - v_j\| \leq \|A_k v_i - A_k v_j\| \leq (1+\varepsilon)\,\|v_i - v_j\|\right).$$

1. Show that $\mathbb{P}(E_{ij}^c) < \delta$, where $E_{ij}^c = \Omega \setminus E_{ij}$.

2. Deduce that
$$\mathbb{P}\left(\bigcap_{i<j} E_{ij}\right) \;\geq\; 1 - \frac{N(N-1)}{2}\,\delta.$$

3. Prove the Johnson–Lindenstrauss theorem.